RFI Report for December 2024

Executive summary

Overall, based on the samples that have been collected, the cross correlation products of the selected observations are minimally, if at all, affected by RFI emissions. The autocorrelation flags of some observations (e.g. subsection a, subsection c, subsection d and subsection f) exhibit relatively weak RFI contamination on the lower part of the band, although their corresponding cross correlations show a decreasing number of affected channels. In the October report, a sub-band (811 - 821 MHz) was shown to be contaminated by an emission from Telkom downlink. Again, in an observation on the 8th of December, it has been found that the same sub-band has been used as evidenced by the RFI emission in the flags shown in subsection b. The strength of the noise suggests that the antennas in the core are the most affected, i.e. it is stronger on short baselines with a fraction of the time flagged about 0.7.

Building on the analysis of autocorrelation and cross-correlation products in the 32k mode, the second segment of the report focuses on cross-correlation RFI behavior across both 32k and 4k modes for November 2024. The hourly analysis (Figure 19 in section 4) shows average RFI levels at 35%, with notable peaks at hours in the early morning, and late evening, with an average RFI percentage of 60%. Figure 19 reveals intense RFI activity in bands such as 544–592 MHz and 910–950 MHz and above 1020 MHz, overlapping with previously identified sources.

1. Introduction

The purpose of this document is to report the current status of RFI on-site. Given the construction work that has been going on over the last couple of months, it is therefore imperative to monitor the construction related activities on site that could potentially generate RFI signals which, in turn, contaminate the science observations. Preferably these reports are to be released on a weekly basis (or every two weeks).

2. Method

The flags data, specifically *ingest_flags*, are considered in these analyses, as they can be used as proxies of RFI frequency occupancy, i.e. RFI emission as a function of frequency channels. The strategy is to select two to three observations per week and report the status of the RFI contamination on site based on the flags of each dataset (of each observation). For each observation

- Autocorrelation flags of a few *scans* corresponding to the *gain calibrator* are selected. This is due to the fact that, at the time of writing, there are some issues with accessing a big chunk of an entire dataset.

- Autocorrelation flags, which are three dimensional arrays, i.e. *time*: *frequency channels*: *correlation products*, are averaged over both time and correlation products.
- Autocorrelation flags become a one dimensional array which is plotted in order to check the RFI occupancy in the frequency channels.

The cross correlation flags are also investigated in order to see how much the RFI emission affects the correlation products which are the main interest in an interferometric observation. The procedure for the cross correlation flags is the same as that of autocorrelation ones. To further go into more detail, the flags corresponding to short and long baselines are split into two parts, and each part is plotted separately. Based on some datasets that were analyzed previously, the U-band observations are the ones that are most affected by the current RFI emissions on site. Hence, the majority of the datasets that will be inspected is in U-band, nevertheless observations in both L and S bands will be looked at too.

In what follows, it is worth noting that the vertical gray shaded areas in all plots denote the known RFI emissions in U-band.

3. Results

a. 2024, 01 December

The autocorrelation, short and long baselines cross correlation flags of the observation collected on the 1st of December are shown in Figures 1, 2 and 3 respectively. Overall the flags seem clean, i.e. they are contaminated by RFI emissions, although a relatively weak RFI, with a fraction of time flagged less than 0.1, is noticed in all three types of flags with varying strength. It appears that the emission is related to DTT Beauford West 634.

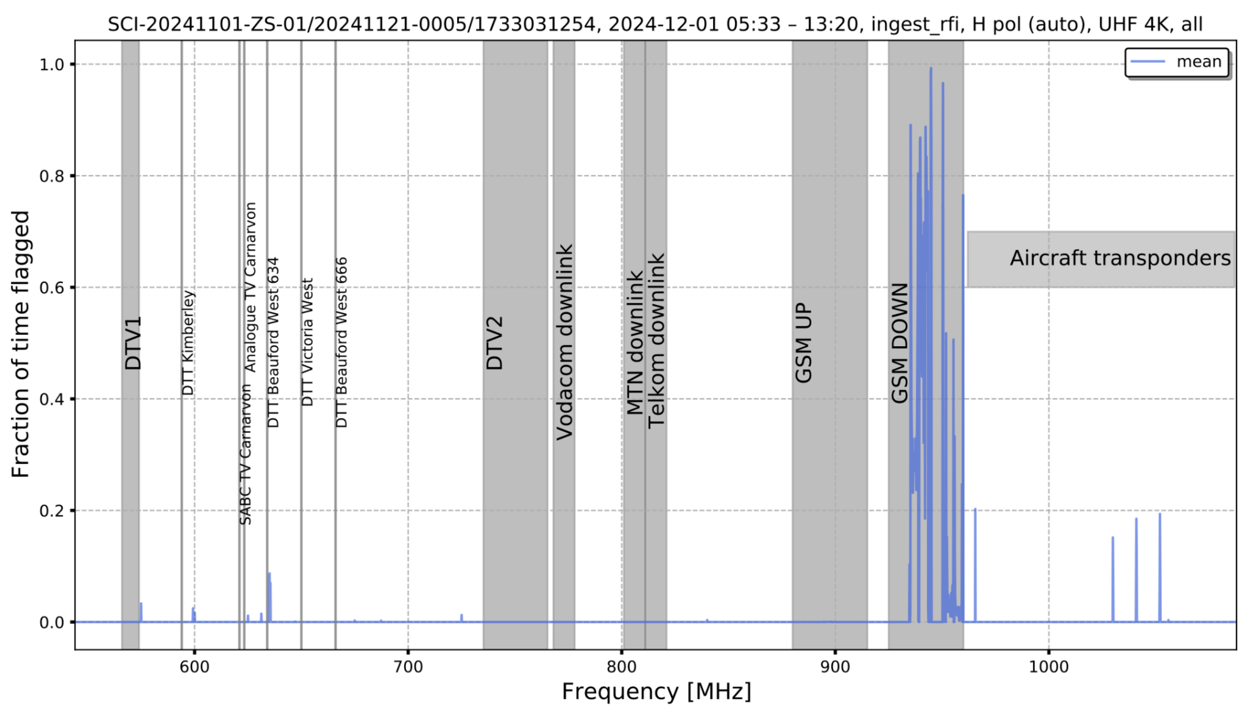

Figure 1: Mean of the autocorrelation flags of SB_ID:20241121-0005 dataset.

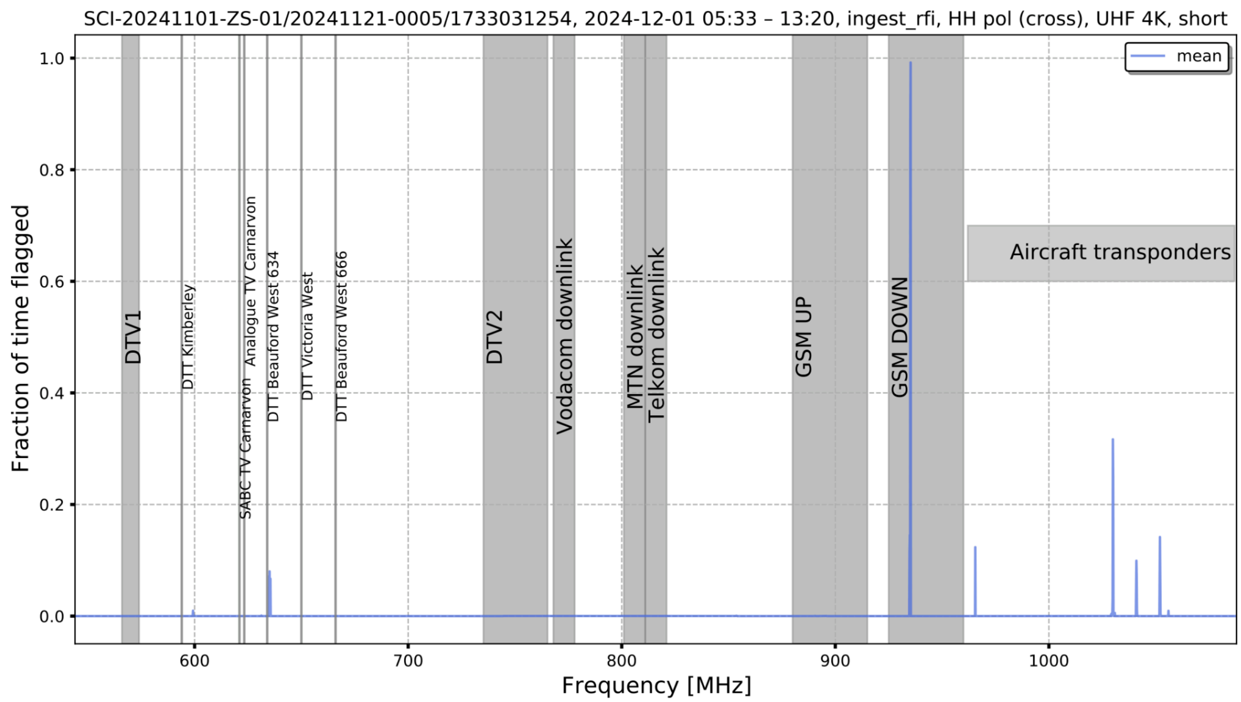

Figure 2: Mean of the cross correlation flags of the short baselines of SB_ID:20241121-0005 dataset.

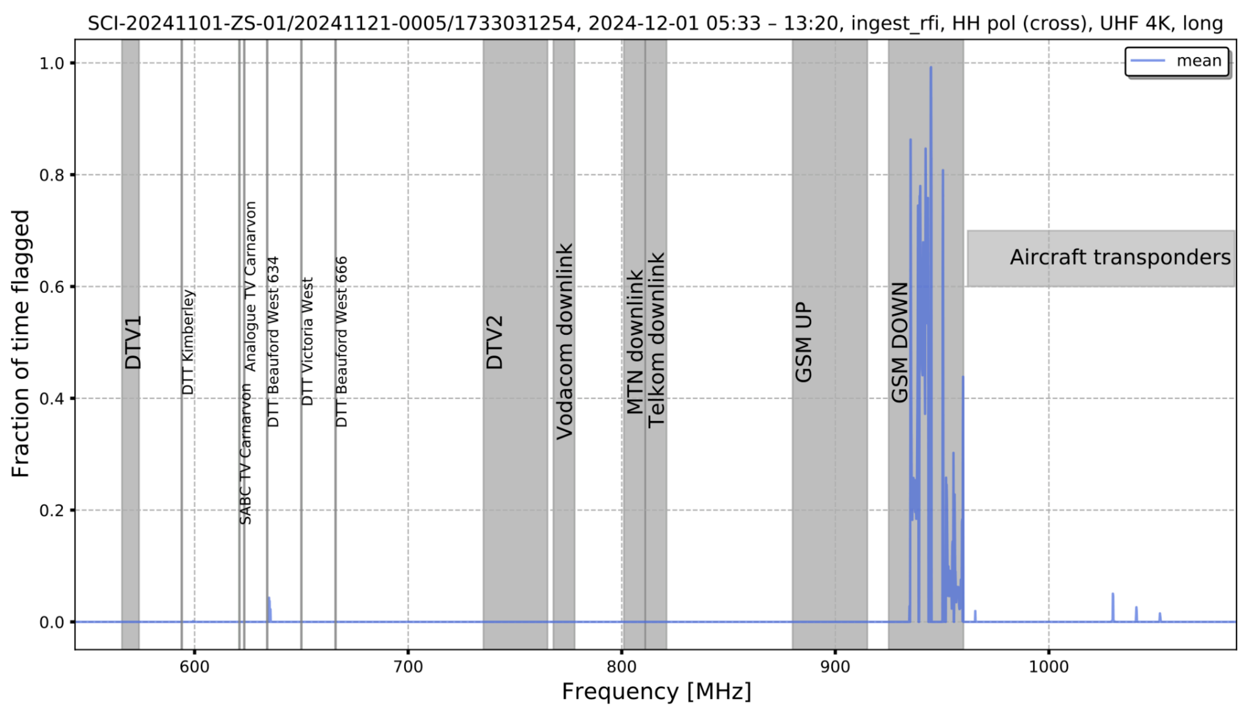

Figure 3: Mean of the cross correlation flags of long baselines of SB_ID:20241121-0005 dataset.

b. 2024, 08 December

It is quite clear from the different flags of the data collected on the 8th of December, as shown in Figures 4, 5, and 6, that the Telkom Downlink still generates RFI. The fact that the contaminant is strongest on the short baselines (as shown in Figure 5) indicates it is the antennas at the core that are most affected. Otherwise, the other frequency channels are visibly clean.

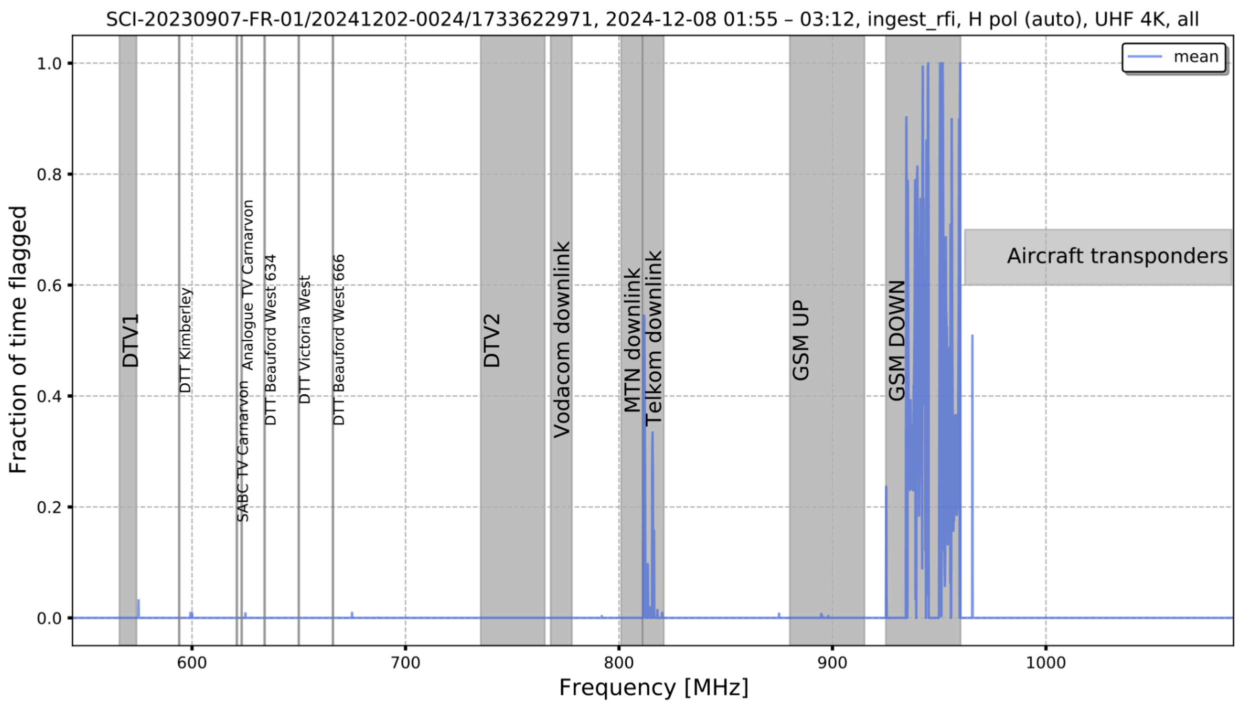

Figure 4: Mean of the autocorrelation flags of SB_ID:20241202-0024 dataset.

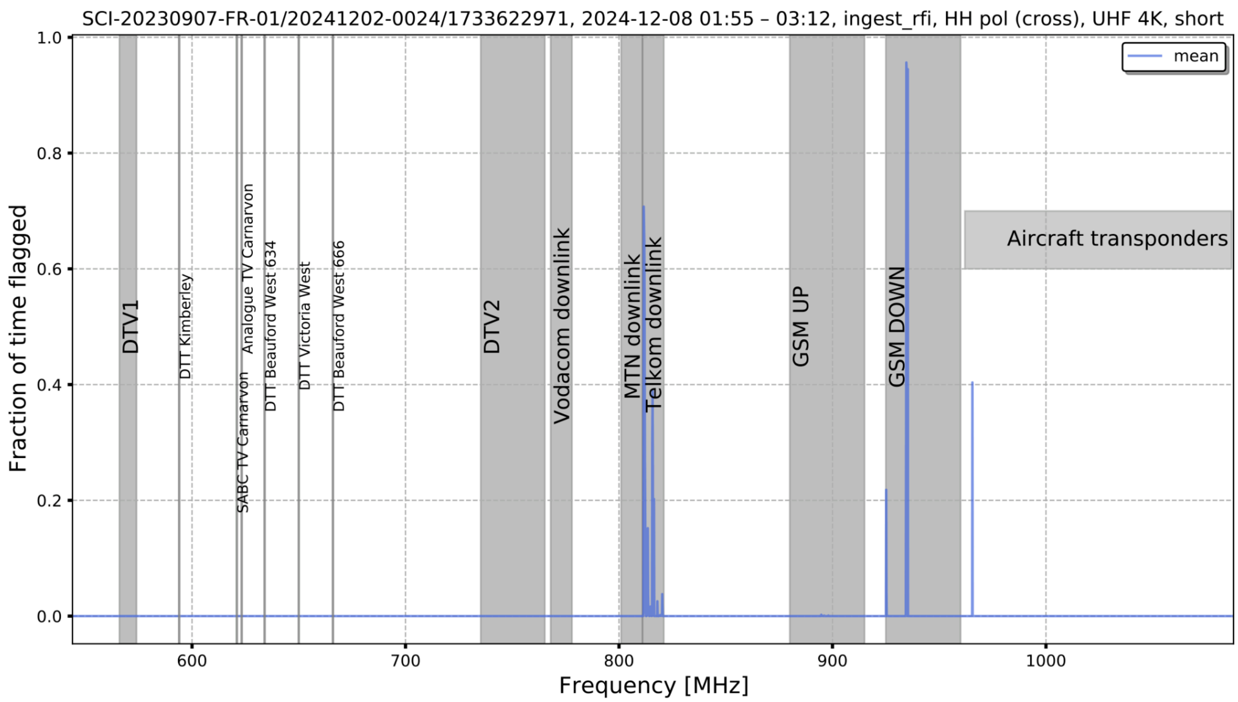

Figure 5: Mean of the cross correlation flags of the short baselines of SB_ID:20241202-0024 dataset.

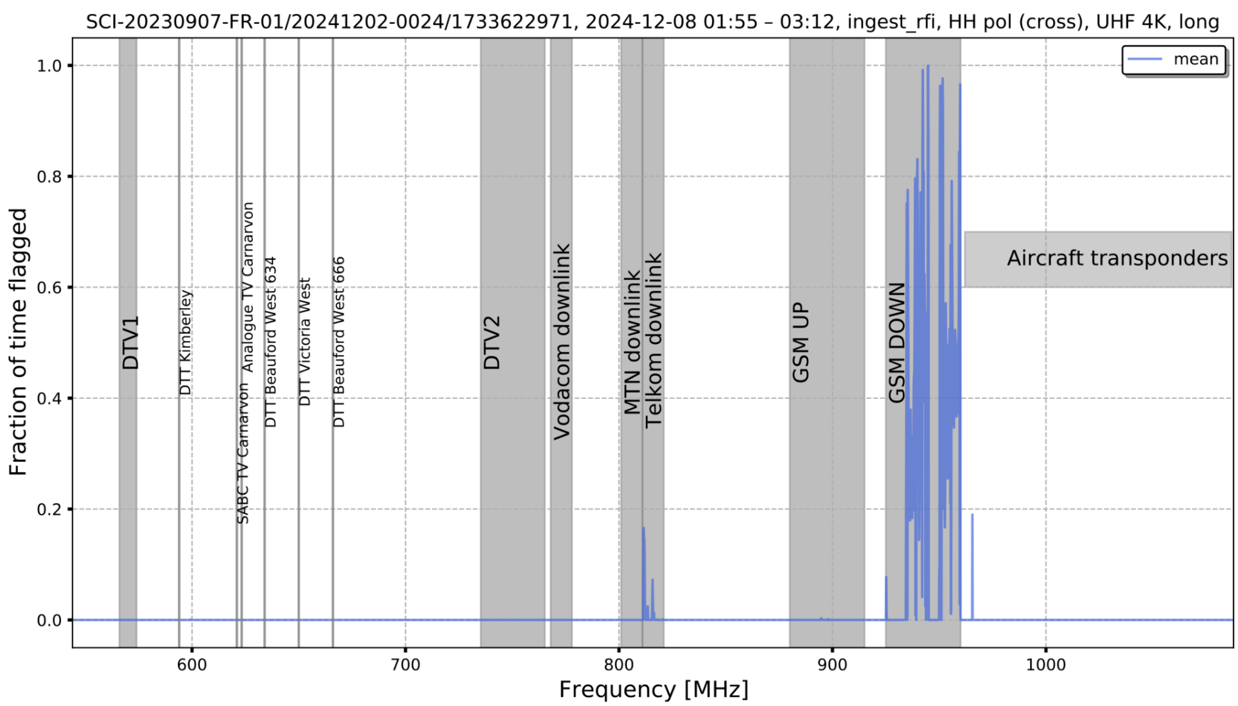

Figure 6: Mean of the cross correlation flags of long baselines of SB_ID:20241202-0024 dataset.

c. 2024, 13 December

The autocorrelation flags of the observation from 13th of December, as presented in Figure 7, exhibit relatively weak RFI on the lower part of the band. The number of contaminated channels decreases in the short baselines cross correlation (see Figure 8) and further decreases in the long baselines correlation as evidenced by Figure 9.

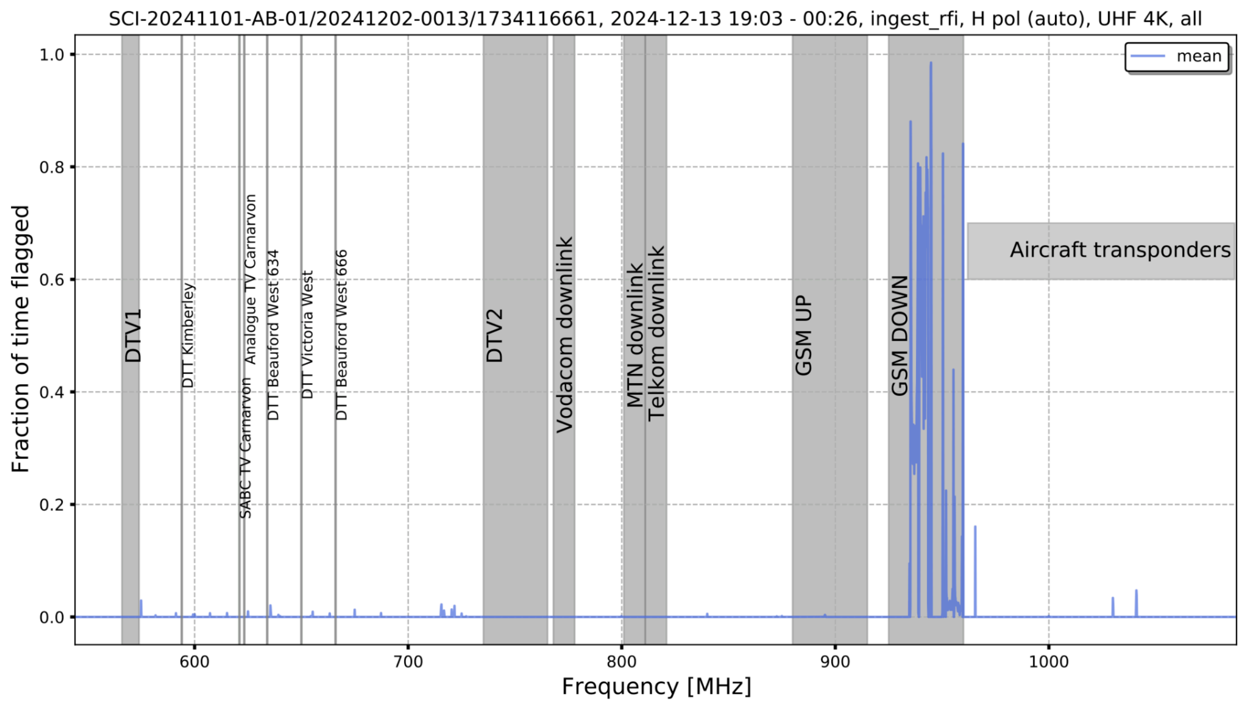

Figure 7: Mean of the autocorrelation flags of SB_ID:20241202-0013 dataset.

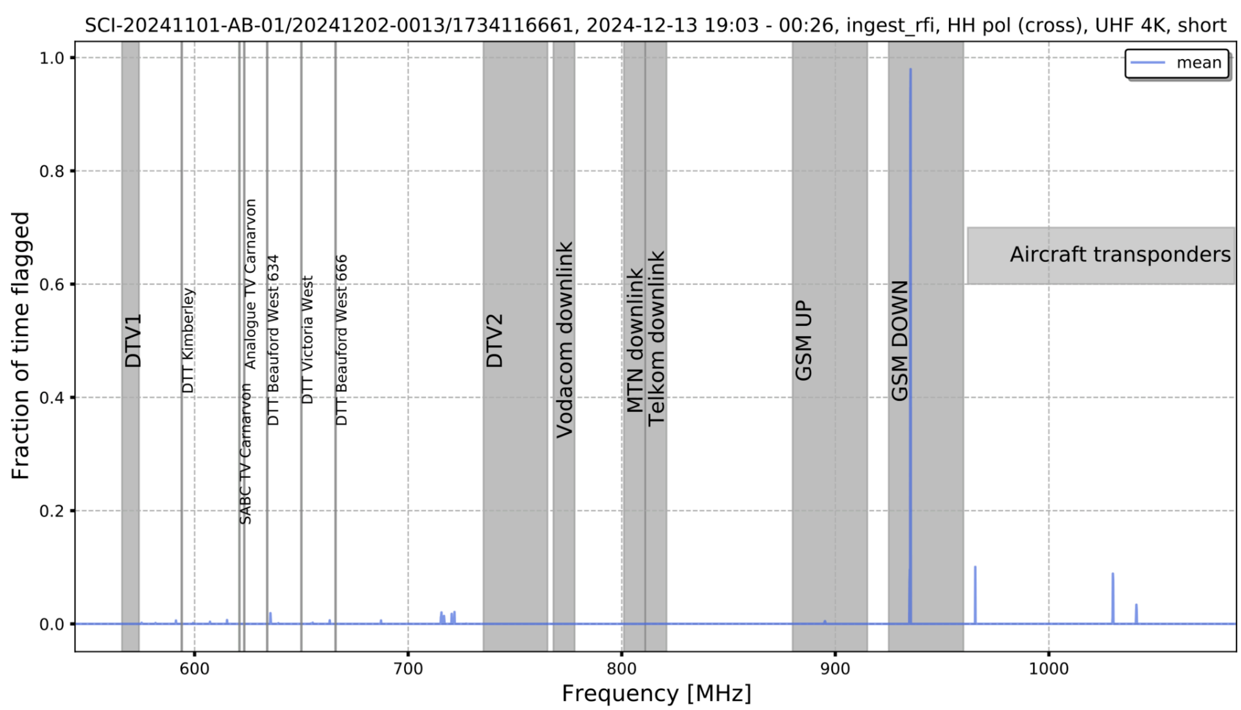

Figure 8: Mean of the cross correlation flags of the short baselines of SB_ID:20241202-0013 dataset.

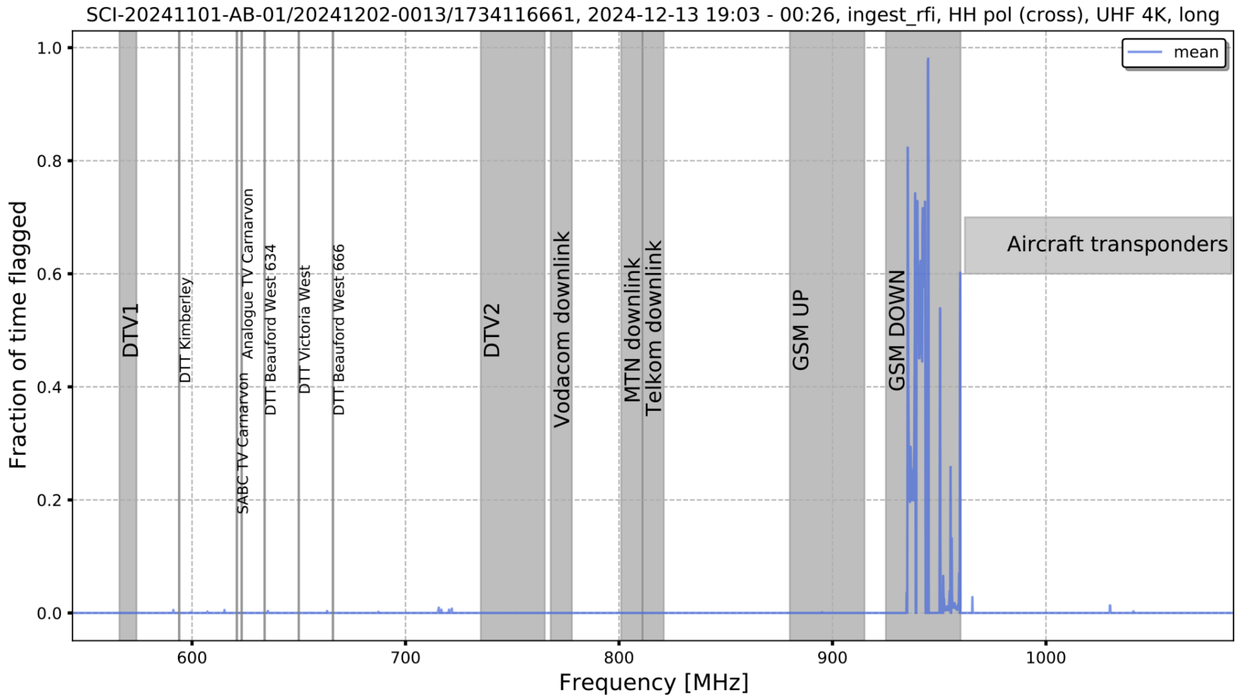

Figure 9: Mean of the cross correlation flags of the long baselines of SB_ID:20241202-0013 dataset.

d. 2024, 21 December

As presented in Figure 10, the RFI contamination in the autocorrelation flags of the data obtained on the 21st of December is relatively low, i.e. a few channels affected by RFI with fraction of time flagged less than 0.1. Both the short and long baselines correlations are cleaner as shown in Figures 11 and 12.

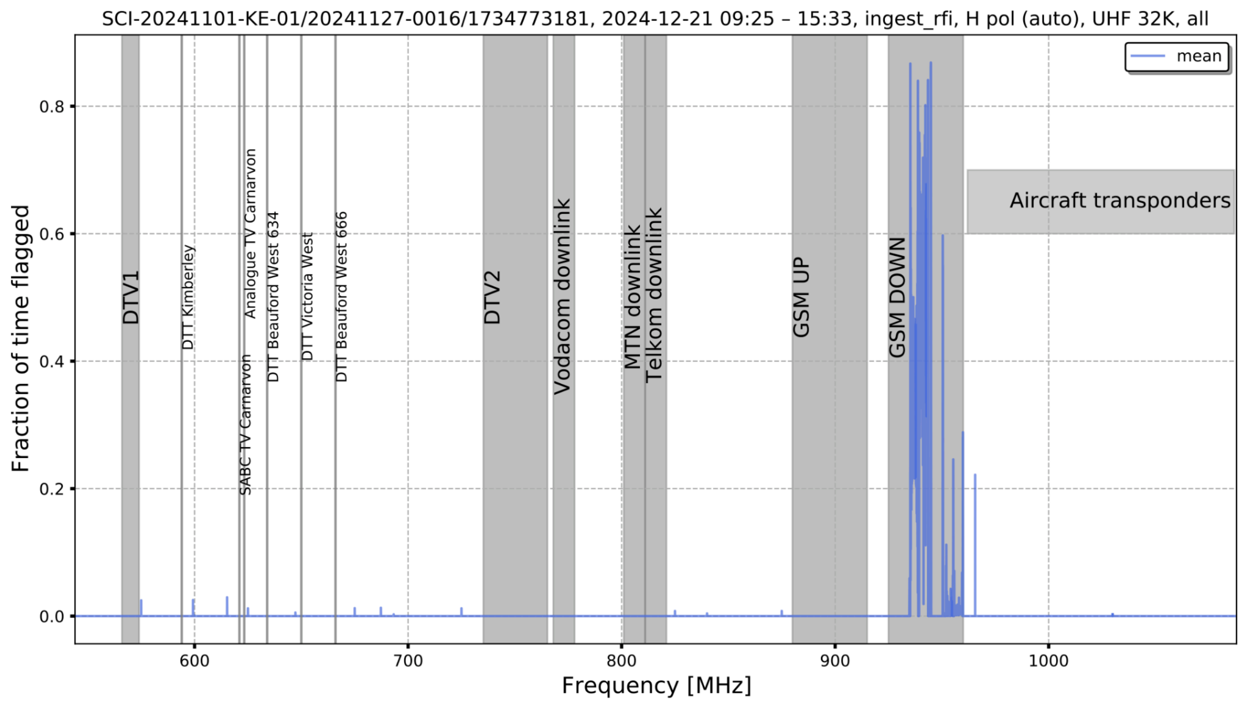

Figure 10: Mean of the autocorrelation flags of SB_ID:20241127-0016 dataset.

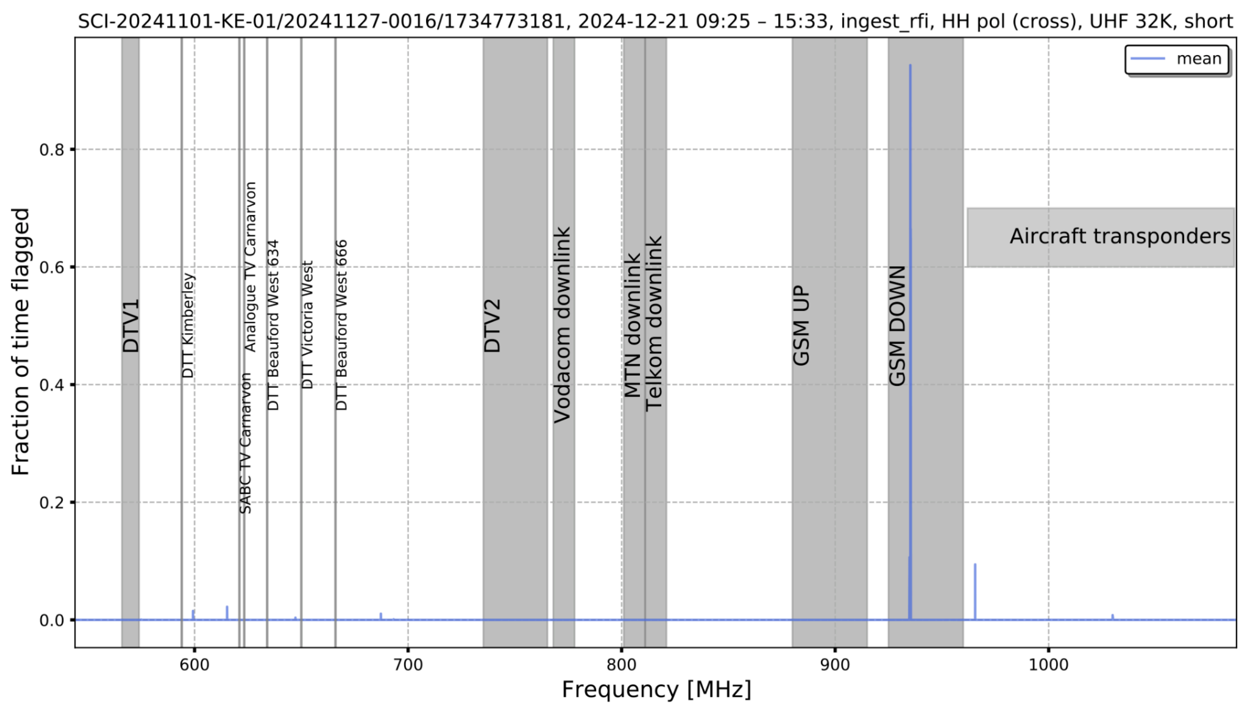

Figure 11: Mean of the cross correlation flags of the short baselines of SB_ID:20241127-0016 dataset.

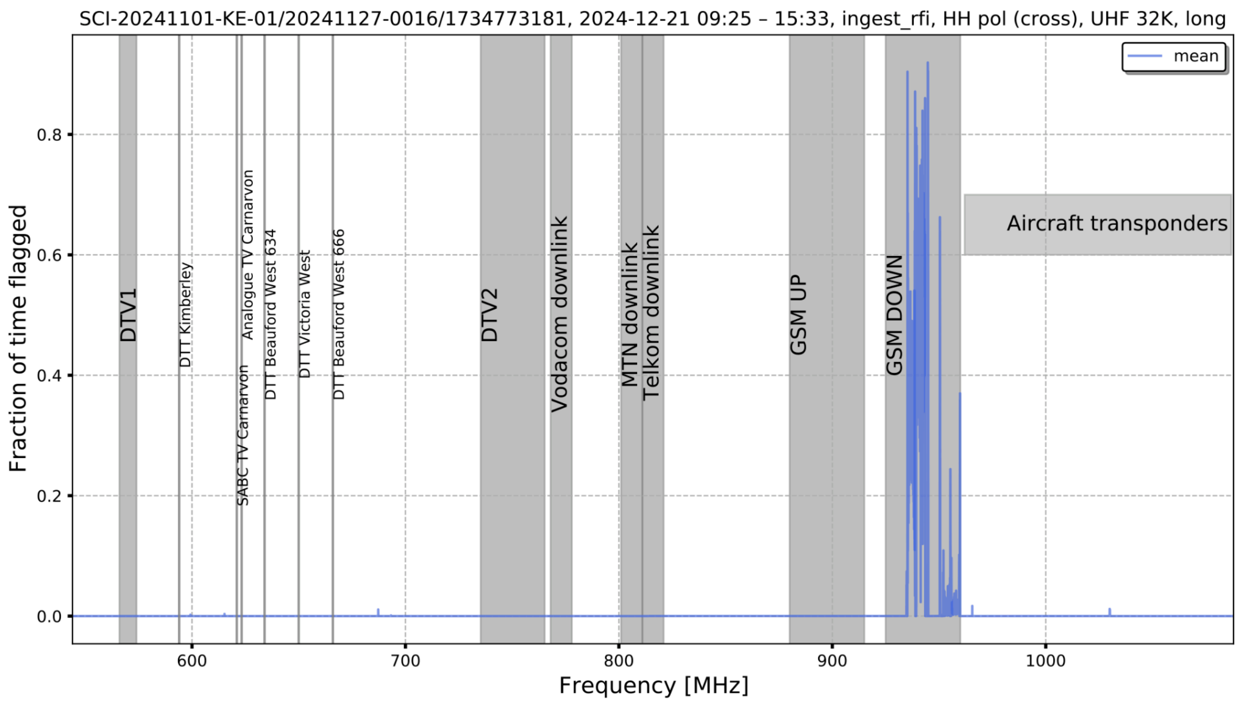

Figure 12: Mean of the cross correlation flags of the long baselines of SB_ID:20241127-0016 dataset.

e. 2024, 24 December

The observation collected on the 24th of December appears to be relatively RFI free, as indicated by its flags in Figures 13, 14 and 15.

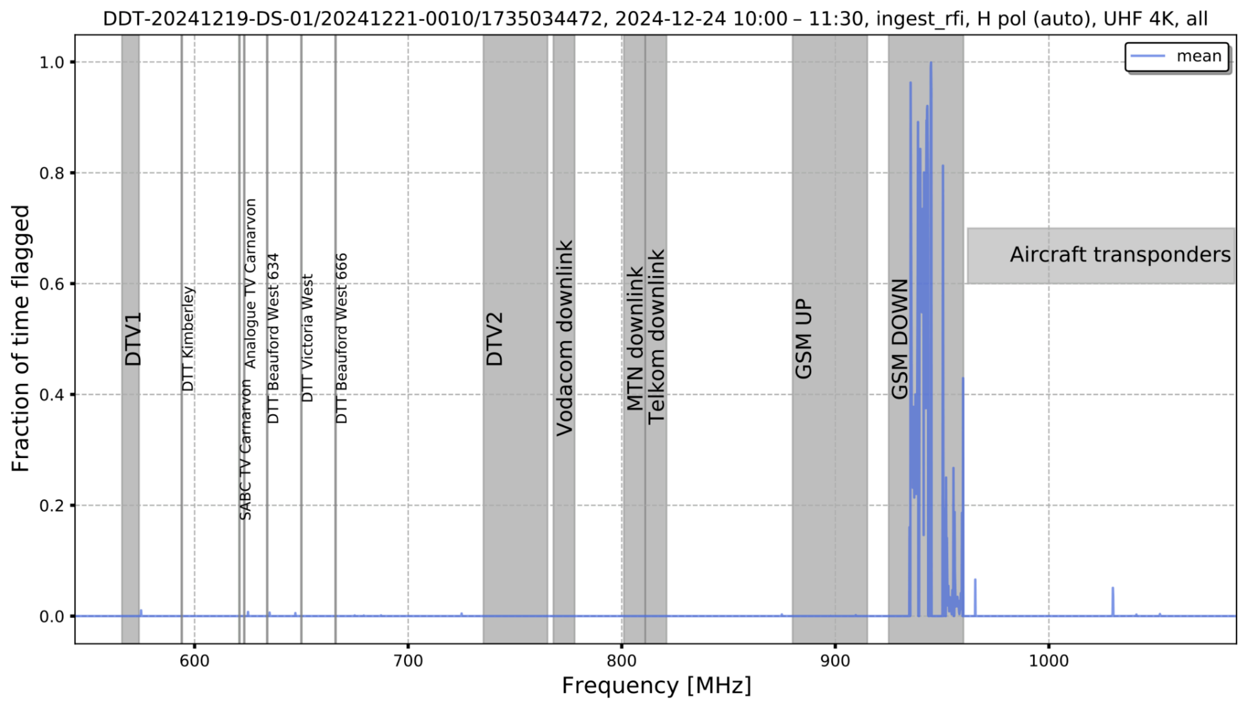

Figure 13: Mean of the cross correlation flags of the long baselines of SB_ID:20241221-0010 dataset.

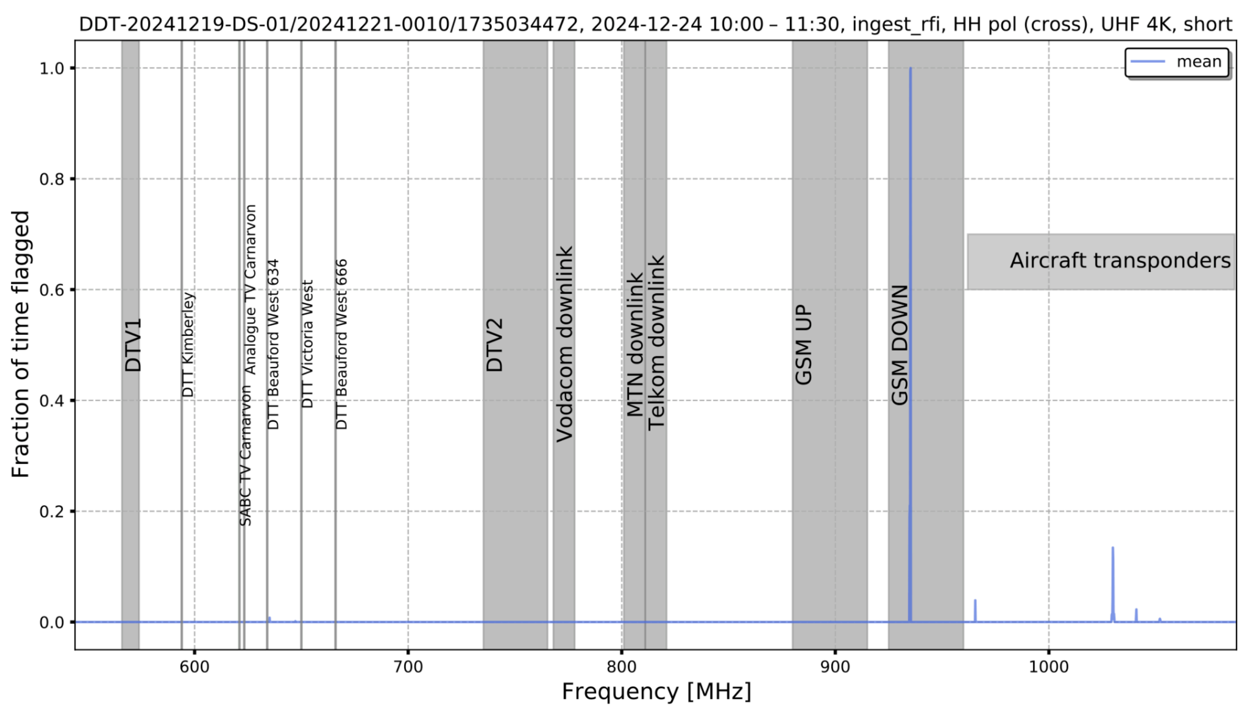

Figure 14: Mean of the cross correlation flags of the short baselines of SB_ID:20241221-0010 dataset.

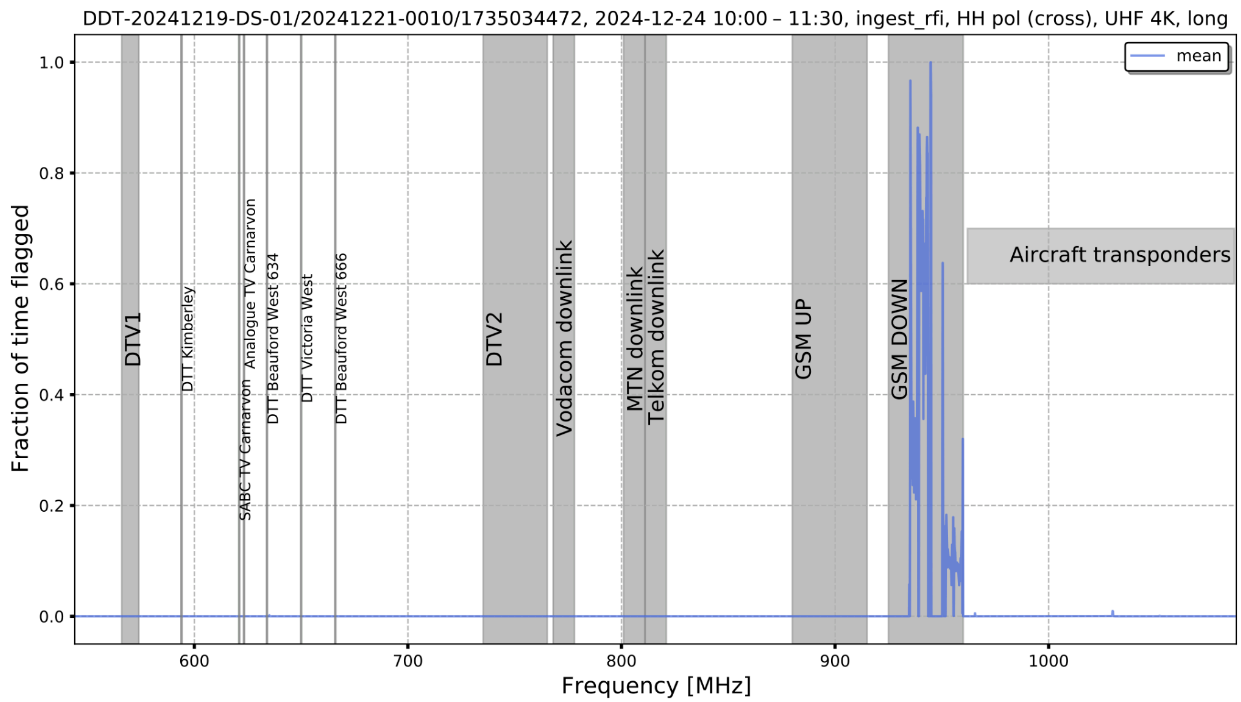

Figure 15: Mean of the cross correlation flags of the long baselines of SB_ID:20241221-0010 dataset.

f. 2024, 29 December

The autocorrelation flags of the data observed on the 29th of December, in Figure 16, exhibit some RFI emissions on the lower part of the band. The cross correlation products (short and long baselines), on the other hand, appear to be clean, as evidenced by Figures 17 and 18.

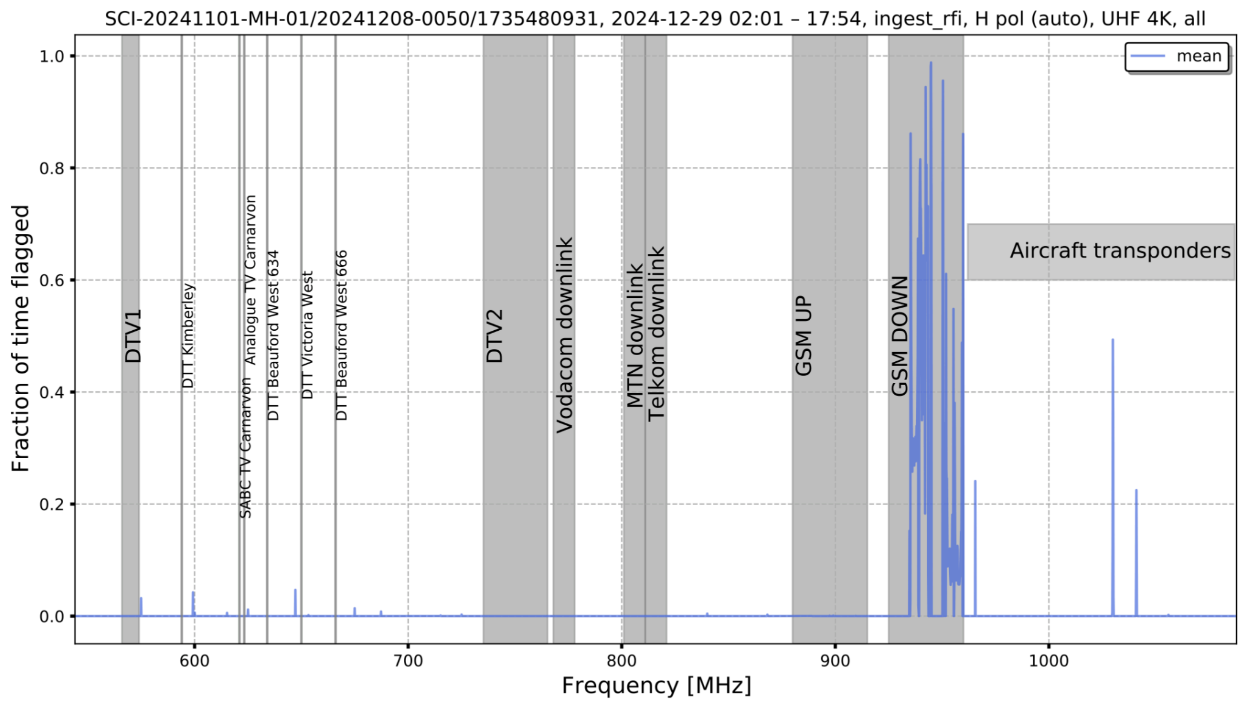

Figure 16: Mean of the cross correlation flags of the long baselines of SB_ID:20241208-0050 dataset.

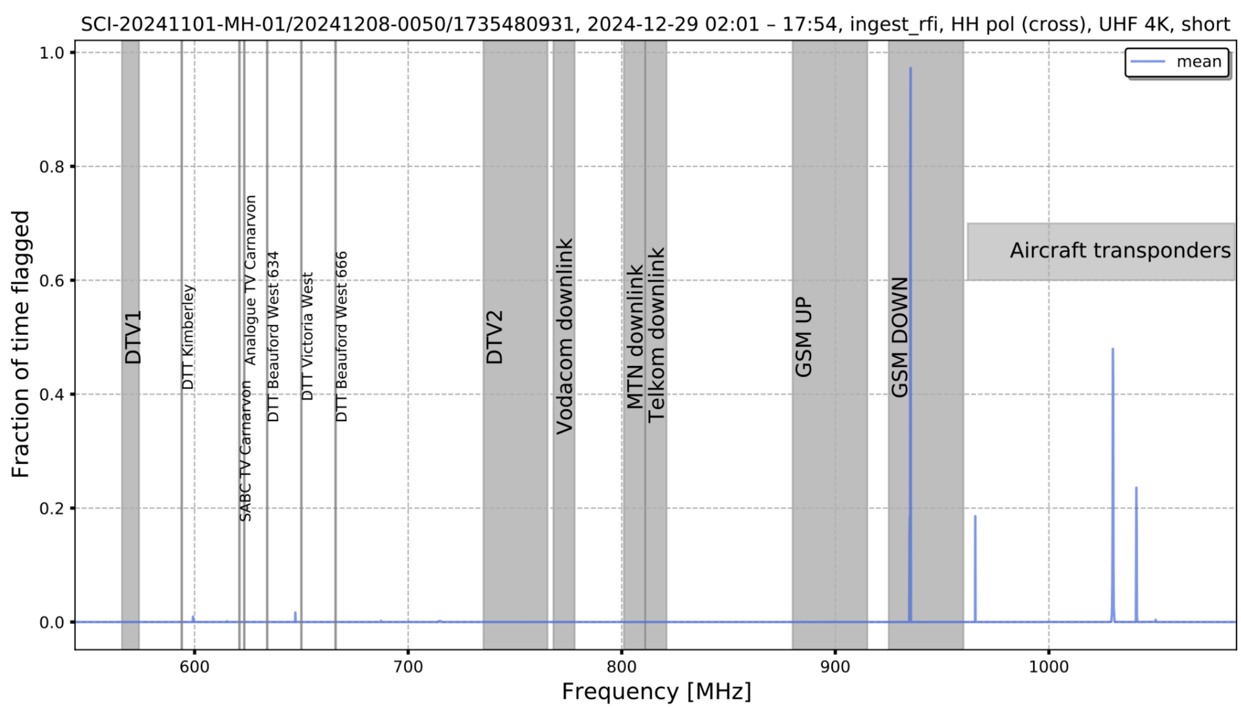

Figure 17: Mean of the cross correlation flags of the short baselines of SB_ID:20241208-0050 dataset.

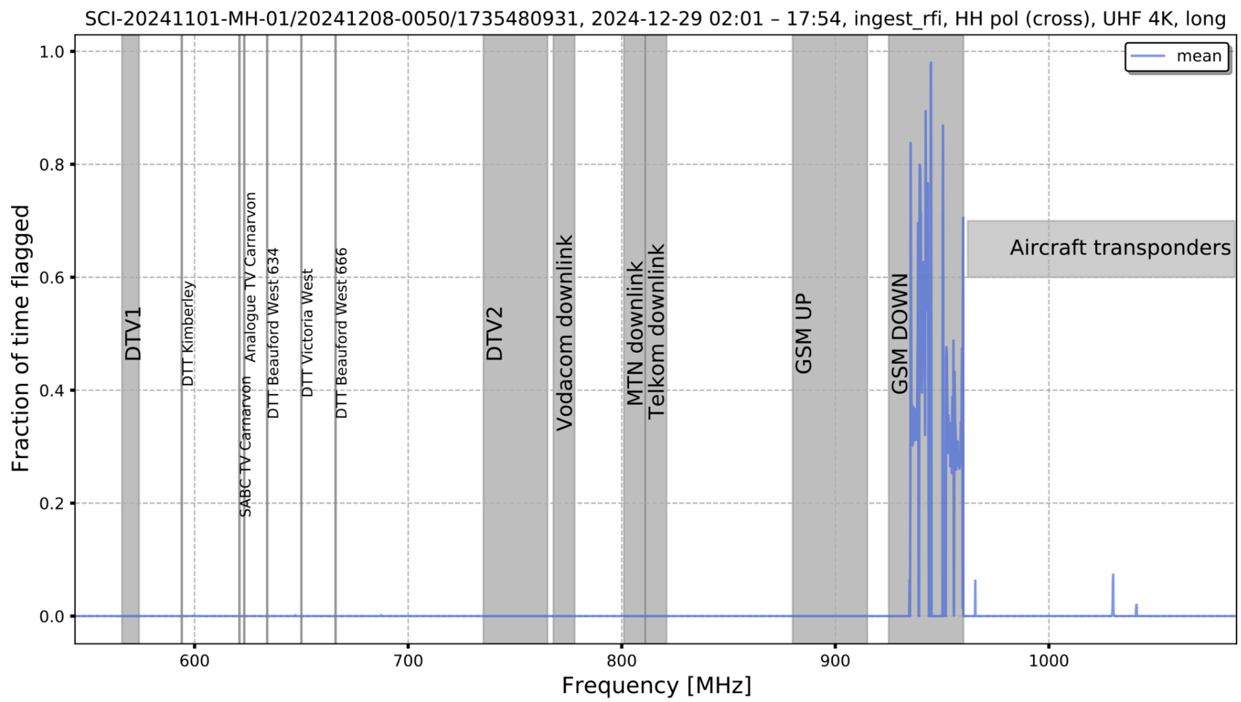

Figure 18: Mean of the cross correlation flags of the long baselines of SB_ID:20241208-0050 dataset.

4. RFI summary for November 2024 at UHF, HH cross polarization

Now we look at the overall RFI for the month of September by concatenating all science observations for the month. Here we select particularly 8s observations, observed at 4k or 32k channel mode. The monitoring and analysis of RFI are presented using the KATHPRFI pipeline [1], highlighting the impact of construction on RFI levels.

KATHPRFI algorithm

The kathprfi essentially processes two numpy arrays, Master and Counter, to update the number of RFI points and observations per voxel which refers to 5-D array of Time, Frequency, Baseline, Elevation and Azimuth. It iterates over chunks of frequency indices, baseline indices, and time indices. For each combination, the algorithm updates the Master array by the corresponding ‘Good flags‘ value and the Counter array by 1. Good flags for a given observation are based on a flag type, e.g. if flag type is calibration RFI then target tags for that observation will be obtained from a complete calibration report. The output of this is an xarray dataset that contains arrays of these two arrays and the dimensions of interest as mentioned above.

Computation of marginal probability

The RFI probability is computed by using two arrays from the output Zarr file obtained from the kathprfi pipeline : ’Master‘ and ‘Counter‘. These arrays represent the number of RFI points and the total number of observations, respectively. The function sums these arrays along a specified dimension to get total counts and then calculates the probability as the ratio of these sums.

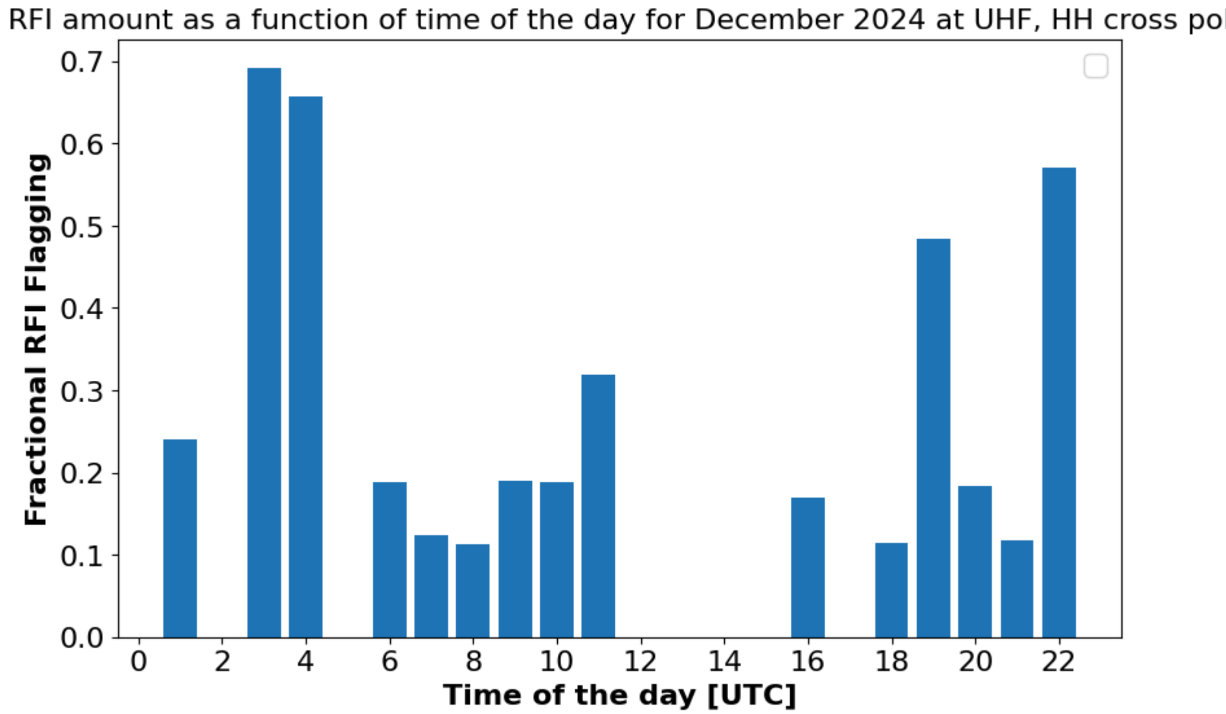

Figure19: The plot above shows the fractional RFI flagging as a function of time of the day in UTC for December 2024 UHF horizontal cross pol observations.

Figure 19 illustrates the hourly behavior of RFI for November 2024, highlighting specific time periods affected by elevated RFI levels. The data reveals that RFI behavior is generally unstable, with an average RFI percentage of 35% throughout the day, although most of the time RFI percentage is relatively low (less than 30%). Notably, RFI levels exceed the daily average during the hours 03:00 UTC, 04:00 UTC, 19:00 UTC and 22:00 UTC. Among these, a particularly high RFI fraction is observed at (03:00 and 04:00) UTC, where the RFI percentage peaks at around 70%. This suggests a period of heightened RFI prevalence. The causes for such elevated RFI levels may include specific construction activities on-site or interference related to a source under observation during that time. To better characterize these anomalies, a detailed analysis of the frequency spectra during these hours is necessary.

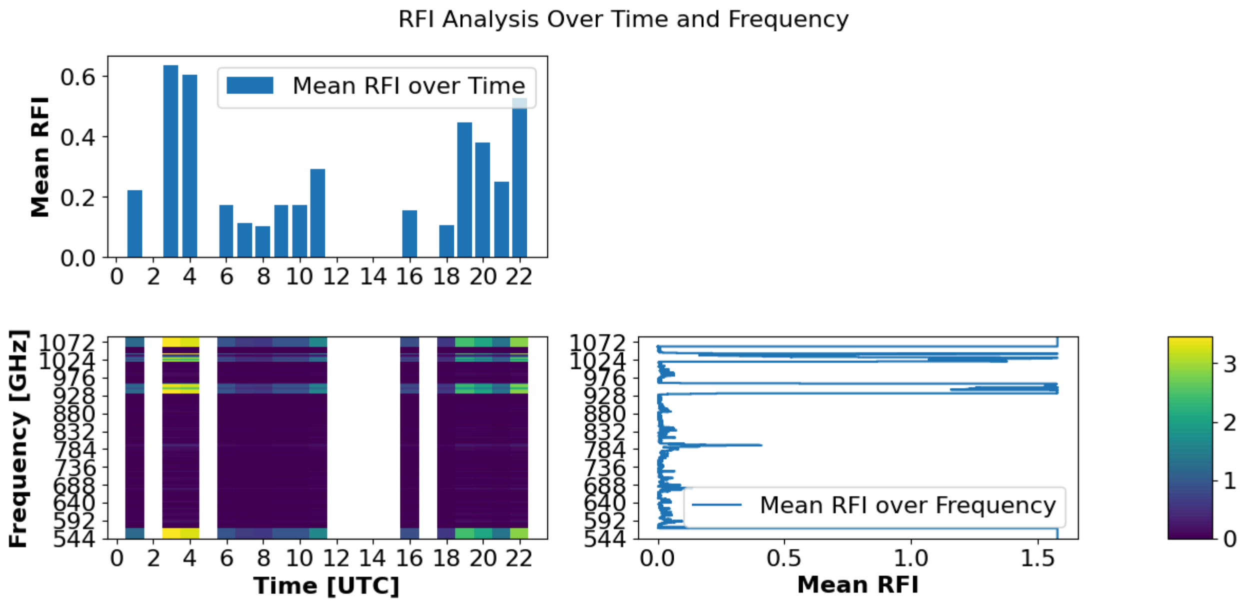

Figure 20: 2-D average RFI occupancy across time and frequency space

Figure 20 presents a two-dimensional view of RFI patterns across time and frequency. The top 1D bar plot displays mean RFI levels over time, consistent with the trends shown in Figure 19. The central 2D heatmap provides a detailed visualization of RFI activity, with white regions indicating missing data or non-monitored frequency intervals. Intense RFI zones are represented by bright yellow or green patches, as per the color bar on the far right. Significant RFI is observed in specific frequency bands, including 544–592 MHz, 910–950 MHz, and above 1000 MHz. The 1D plot on the right highlights average RFI levels across frequencies, where noticeable spikes represent strong interference at specific frequency ranges.

We recognize known RFI sources in some bands, such as DTV1 (544–592 MHz), GSM downlink/uplink (910–950) MHz and above 1020 MHz.